

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May - June 2023

MA1003-1 – DIFFERENTIAL EQUATIONS AND LAPLACE TRANSFORMS

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- $xy \frac{dy}{dx} + 2x^3 y \frac{d^2y}{dx^2} = 0$ is a differential equation of order
 A) 1
 B) 2
 C) 4
 D) 3
- The differential equation $\frac{dy}{dx} + (5y) = x^4 y^7$ is
 A) Linear
 B) Exact
 C) Bernoulli's
 D) Of order 2
- The differential equation $y = 2px + y^2 p^3$ is
 A) Linear
 B) Solvable for p
 C) Solvable for x
 D) Solvable for y
- Solution of $5 \frac{di}{dt} + i = t$ is
 A) $ie^{\frac{t}{5}} = 5te^{\frac{t}{5}} - 25e^{\frac{t}{5}} + c$
 B) $ie^{\frac{t}{5}} = 5te^{\frac{t}{5}} - 25e^{\frac{t}{5}} + c$
 C) $ie^{\frac{t}{5}} = te^{\frac{t}{5}} - 5e^{\frac{t}{5}} + c$
 D) $ie^{\frac{t}{5}} = 5te^{\frac{t}{5}} - 5e^{\frac{t}{5}} + c$
- The complementary function of the differential equation $x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} = x^4$ is
 A) $y_c = c_1 + c_2 x^2$
 B) $y_c = c_1 + c_2 e^t$
 C) $y_c = c_1 + c_2 e^{4t}$
 D) $m^2 - 2m = 0$
- Auxiliary equation of $x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} - y = 2x \log x$ is
 A) $m^2 - 2m - 1 = 0$
 B) $m^2 - 3m - 1 = 0$
 C) $m^2 - 3m - y = 0$
 D) $m^2 - 3m - y = 2x \log x$
- The absolute value of the Wronskian of e^{3x} and e^x is
 A) $2e^{4x}$
 B) $-2e^{4x}$
 C) $2e^{4x}$
 D) $-2e^{-4x}$
- The solution of the homogeneous equation $(6D^3 + D^2)y = 0$ is
 A) $y = C_1 + C_2 x + C_3 e^{-\frac{1}{6}x}$
 B) $y = (C_1 + C_2 x)e + C_3 e^{-\frac{1}{6}x}$
 C) $y = C_1 + C_2 x + C_3 e^{\frac{1}{6}x}$
 D) $y = (C_1 + C_2 x)e^x + C_3 e^{-\frac{1}{6}x}$
- $L\{\tan(\frac{\pi}{4})\} =$
 A) $\sqrt{2}s$
 B) $\frac{1}{\sqrt{2}s}$
 C) $\frac{\sqrt{2}}{s}$
 D) $\frac{1}{s}$
- $L\{10t\} =$
 A) $\frac{10}{s}$
 B) $\frac{10}{s^2}$
 C) $\frac{20}{s}$
 D) $10s$
- $L\{u(t+2)\} =$
 A) $\frac{e^{-2s}}{s^2}$
 B) e^{2s}
 C) $\frac{e^{2s}}{s}$
 D) $\frac{e^{-2s}}{s}$

12. $L\{5^t\} =$
 A) $\frac{1}{5s}$ B) $\frac{1}{s-\log 5}$
 C) $\frac{1}{s-5}$ D) s
13. $L^{-1}\left\{\frac{1}{s}\right\} =$
 A) 1 B) t
 C) t^2 D) s
14. $L^{-1}\left\{\frac{1}{(s-3)^2}\right\}$ is
 A) $\frac{t^2}{2}e^{-3t}$ B) te^{-3t}
 C) te^{3t} D) $\frac{t}{2}e^{3t}$
15. $L^{-1}\left\{\frac{e^{-3s}}{s^2}\right\} =$
 A) $(t+3)u(t+3)$ B) $(t-3)u(t-3)$
 C) $(t+3)u(t-3)$ D) none of these
16. $L^{-1}\left\{\frac{s-a}{(s-a)^2-b^2}\right\} =$
 A) $e^{at}\cosh bt$ B) $e^{-at}\cosh bt$
 C) $\frac{e^{-at}\cosh bt}{b}$ D) $\frac{e^{at}\cosh bt}{b}$
17. One dimensional heat equation is
 A) $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ B) $\frac{\partial u}{\partial t} = c^2 \frac{\partial u}{\partial x}$
 C) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ D) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial u}{\partial x}$
18. $\cos(5x+6y) \frac{\partial^3 u}{\partial x^3} + \frac{\partial^2 u}{\partial x^2} = 2xy$ is a partial differential of degree
 A) 1 B) 2
 C) 3 D) 4
19. The partial differential equation formed from $z = (c+x)^2 + y$ is
 A) $z = p^2 + y$ B) $z = q^2 + y$
 C) $z = \frac{1}{4}p^2 + y$ D) $z = \frac{1}{4}q^2 + y$
20. One dimensional wave equation is
 A) $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ B) $\frac{\partial u}{\partial x} = c^2 \frac{\partial^2 u}{\partial t^2}$
 C) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ D) $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x}$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1. a)	Solve $r \sin \theta - \cos \theta \frac{dr}{d\theta} = r^2$.	6	L2	1	2
b)	Using the method of variation of parameters, solve $(D^2 - 4D + 4)y = \frac{e^{2x}}{x}$	6	L3	2	2
c)	Solve $p = \sin(y - xp)$, also find its singular solution.	4	L2	1	2
2. a)	If 30% of the radioactive substance disappeared in 10 days, find how long will it take for 90% of it to disappear.	6	L3	1	2
b)	Solve $(D^3 - D)y = 2x + 1 + 4\cos x + 2e^x$.	6	L2	2	2
c)	Solve $(D^2 + 3D - 4)y = 12e^{2x}$.	4	L2	2	2
3. a)	Find orthogonal trajectories of $r^n = a \sin n\theta$.	6	L2	1	2
b)	Solve $x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} - 12y = x^3$.	6	L3	2	2
c)	Solve $\frac{d^2 x}{dt^2} + 5 \frac{dx}{dt} + 6x = 0$ given $x(0) = 0, \frac{dx}{dt}(0) = 15$	4	L2	2	2
Unit - II					
4. a)	If $L\{f(t)\} = F(s)$ then prove that $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}(F(s))$.	6	L3	3	2
b)	Use partial fraction method to find inverse Laplace transform of $\frac{2s+3}{(s-1)(s+2)^2}$	6	L2	4	2

- c) Find Laplace transform of $e^{-3t}(2\cos 5t - 3\sin 5t)$. 4 L2 3 2
5. a) If $f(t)$ is a periodic function with period T , then prove that

$$L\{f(t)\} = \frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}}$$
 6 L3 3 2
- b) Using Convolution theorem find the inverse Laplace transform of $\frac{1}{s^2(s-1)}$. 6 L2 4 2
- c) Find the Laplace Transform of $\int_0^t e^{-t} \cos t dt$. 4 L2 3 2
6. a) Rewrite the following function using unit step function and find its Laplace transform $f(t) = \begin{cases} \sin t, & 0 < t \leq \frac{\pi}{2} \\ \cos t, & t > \frac{\pi}{2} \end{cases}$. 6 L2 3 2
- b) Using Laplace transform method, solve the differential equation $x''(t) + 4x'(t) + 4x(t) = 4e^{-2t}$; $x(0) = -1, x'(0) = 4$. 6 L3 4 2
- c) Find $f(t)$ if $L\{f(t)\} = \frac{e^{-3s}}{(s-4)^2}$. 4 L2 4 2

Unit - III

7. a) Form the partial differential equations by eliminating the arbitrary constants/ functions from
 i. $z = xy + y\sqrt{x^2 - a^2} + b$
 ii. $z = f(x^2 - y^2)$ 6 L3 5 2
- b) Derive one dimensional heat equation in the form $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ 6 L2 5 1
- c) Solve the PDE $\frac{\partial^2 z}{\partial x \partial y} = \cos(2x + 3y)$ by direct integration method. 4 L2 5 2
8. a) Solve $y^3 \frac{\partial z}{\partial x} + x^2 \frac{\partial z}{\partial y} = 0$ by the method of separation of variables. 6 L3 5 2
- b) Solve $(x^2 - y^2 - z^2)p + 2xyq = 2xz$. 6 L2 5 2
- c) Solve $zpq = p + q$. 4 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
